

Peripheral Inflammation and Physiological Hallmarks of Alzheimer's Disease

Joshua Keene^a, Bhairavi Srinageshwar^a, Swathi Suresh^a, Rebekah Stanley^a, Julien Rossignol^b, Gary Dunbar^a.

a: Neuroscience program, Central Michigan University.

b: College of Medicine, Central Michigan University.

Introduction: Alzheimer's disease (AD) is a prevalent neurodegenerative disease that causes severe cognitive decline and behavioral deficits. There are currently no effective preventative interventions available and no significant treatment options currently exist. Multiple variables likely contribute to the cause and/or progression of this disease.

The major hallmarks of AD pathology include increased production amyloid-beta (A β) and tau proteins. Inflammation also plays an important role in AD—an increase in pro-inflammatory cytokines and immune cells occurs. The relationship between inflammation and immune cell dysfunction is not yet fully elucidated.

Methods: Two mouse models of AD will be used to determine the relationship between inflammation and A β and tau protein deposition and clearance. The 3xTg mouse model includes mutations (transgenic, homozygous)—producing abnormal tau and A β quantity and function. The PS19 mouse model includes only tau abnormalities in mutated (knock-in, hemizygous) positive mice. Mice will receive (**Fig. 2**) either mammalian ringers (placebo group) or infection mimetics starting during the prodromal (early observable symptoms) stage of AD. Mice will receive behavioral testing to determine potential differences in cognitive abilities. They will then be humanely euthanized during the later stage of AD-related pathology (at 10 months old for PS19 mice, and 12 months old for 3xTg mice, **Fig. 3**).

Brain tissue will be sectioned, and immunohistochemistry will be performed to measure biomarkers related to AD pathology—tau, A β , microglia, and astroglia. This stained tissue will then be imaged and antibody quantity will be measured and analyzed to determine the relationship between inflammation and hallmark symptoms of AD.

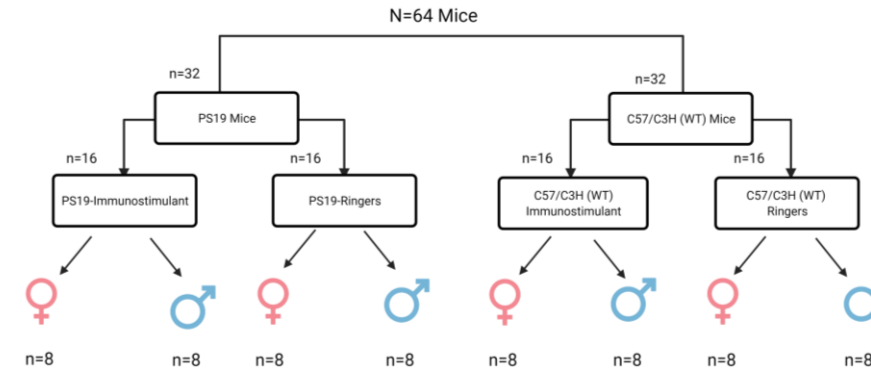


Figure 2: Experimental setup for the PS19 mice and their relevant controls. The same setup (albeit with different mouse strains) will be used for the 3xTg mice.

Figure created using BioRender.com

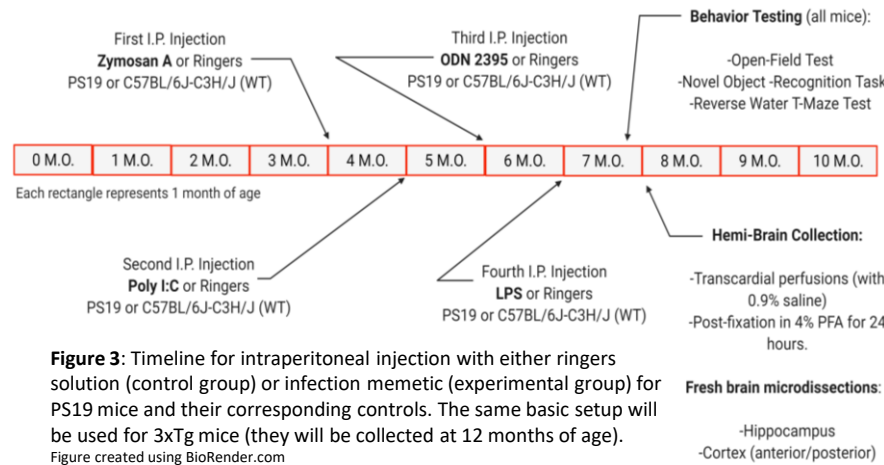
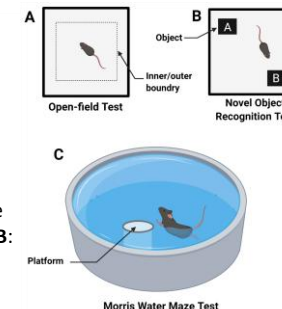


Figure 3: Timeline for intraperitoneal injection with either ringers solution (control group) or infection mimetic (experimental group) for PS19 mice and their corresponding controls. The same basic setup will be used for 3xTg mice (they will be collected at 12 months of age).

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Figure 3: Behavioral testing each mouse will undergo during the latter stages of AD-related pathology, including, A: Open-field, B: novel object recognition task, and C: Morris water maze.

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Hypothesis: Inflammation contributes to the onset and progression of pathological conditions characteristic of AD. The precise relationship between the inflammatory conditions that are present in AD, and the underlying mechanisms driving neurodegeneration have yet to be elucidated. This study will investigate the relationship between infection mimetics and AD pathology—focusing on A β and tau dysfunction and aggregation.

Aim 1: Determine the relationship between infection mimetic immunostimulation and tau pathology using the PS19 mouse model of AD.

I.P. injections will be administered to PS19 mice during the early symptomatic stage of tau pathology. Four immunostimulants will be administered, beginning at 4 months of age (**Fig 2.**). Injections will induce a systemic and prolonged immune response—allowing us to establish the relationship between immune system activation and tau pathology symptomatic of AD.

Aim 2: Determine the relationship between infection mimetic immunostimulation and A β and tau pathology using the 3xTg mouse model of AD.

I.P. injections will be administered to 3xTg mice during the early symptomatic stage of tau and A β pathology. Four immunostimulants will be administered, beginning at 8 months of age. Injections will induce a systemic and prolonged immune response—allowing us to establish the relationship between immune system activation and pathological characteristics of A β and tau dysfunction and aggregation symptomatic of AD.

Rationale: Mouse models of AD induce A β and tau pathology that approximate symptoms characteristic of human AD. These models enable researchers the ability to study correlations and causal mechanisms involved in the etiology and progression of AD. Multiple variables likely contribute to the pathological symptoms relevant to this disease. However, the relationship between pathological hallmarks and immune system function are still poorly understood. This is exemplified by the paradoxical and confounding results obtained from human clinical trials targeting inflammation. Human subjects exposed to anti-inflammatory pharmacological interventions have frequently displayed no significant improvement in cognitive function, nor attenuation of the physiological abnormalities present in AD. Therefore, further investigation into the relationship between inflammatory stress and AD pathology is needed